

FAERS Safety Signal Detection & Disproportionality Analysis

The third-quarter 2020 FAERS data were cleaned and merged with zero record loss, achieving a 100% referential match across demographics, medications, and adverse events. The finalized dataset contains 431,667 unique patient reports, with all duplicate submissions and follow-ups successfully removed. To prevent inflated counts, 347,237 unnecessary drug entries such as multiple doses of the same drug in a single visit were removed. Finally, by filtering exclusively for Primary Suspect (PS) medications, we paired the data with reported reactions to create 1,451,545 drug-event pairs, establishing an accurate baseline for signal detection.

Methodology:

A four-stage data preparation pipeline was developed to transform raw FAERS files containing demographic (DEMO), medication exposure (DRUG), and adverse reaction (REAC) data into a clean, statistically powered signal-detection dataset:

Stage 1: Linking & Deduplication:

First, the encoding format of the data was verified. Next, a complete 1:1 referential match on primaryid was confirmed, with zero dependent records across the three files. To eliminate repeat cases, reports were sorted chronologically by caseid and fda_dt, keeping only the latest regulatory submission for each distinct patient event. Lastly, within-case deduplication removed 347,237 identical drug listings and unwanted reaction terms. This prevents repeated dosages or duplicate reporting from artificially inflating event frequencies. Normalization of drug names was also done.

Stage 2: Primary Suspect Filtering:

The next step focused on filtering the deduplicated cohort by restricting medication exposures to therapeutic agents demoted as **Primary Suspects (role_cod = 'PS')**. Concomitant (C) and interacting (I) medications were excluded to minimise background noise arising from multidrug treatment regimens. Separating the therapeutic agent in the reported adverse event enhances the specificity of exposure assessment and strengthens the assignment of clinical outcomes to the suspected drug.

Stage 3: Vectorized 2 X 2 Contingency Computation:

The filtered drug records were joined with reported adverse reactions (PT) to generate every observed drug-event pair, immediately filtering for pairs with at least three co-occurrences ($\text{age} \geq 3$) to eliminate single-event noise. Rather than relying on computationally heavy loops, the pipeline leveraged marginal totals (N, total drug reports, and total reaction reports) to derive the remaining contingency cells (b, c, and d) via vectorized matrix operations, enabling instantaneous evaluation across millions of records.

Stage 4: Statistical Signal Screening:

Using the derived contingency counts main disproportionality metrics were calculated: the Proportional Reporting Ratio (PRR), Reporting Odds Ratio (ROR) with 95% log-transformed confidence intervals, and closed-form χ^2 test statistics (with Fisher's exact test applied to sparse cells). Potential safety signals were then flagged using standard Evans criteria ($a \geq 3$, $PRR \geq$

2, and $\chi^2 \geq 4$), systematically isolating true disproportionate reporting signals from expected background rates.

For each drug-ADR combination, a 2×2 contingency table was constructed:

	ADR present	ADR absent
Drug present	a	b
Drug absent	c	d

The following disproportionality measures were calculated:

- Proportional Reporting Ratio (PRR)
- Reporting Odds Ratio (ROR)
- 95% confidence interval for ROR
- Chi-square statistic
- Fisher's exact test for small-count combinations

Candidate signals were screened using the assignment criteria of $a \geq 3$, $PRR \geq 2$, and $\chi^2 \geq 4$.

2. Results

Disproportionality analysis identified multiple potential drug-ADR signals satisfying the predefined criteria of $a \geq 3$, $PRR \geq 2$, and $\chi^2 \geq 4$. The highest PRR values were observed for **3TC (lamivudine)** with **jaundice cholestatic** ($PRR = 3340.78$), **hypomania** ($PRR = 3042.50$), and **mitochondrial toxicity** ($PRR = 2148.94$). Other notable signals included 3TC with hepatotoxicity and blood bilirubin increased, as well as hematological reactions associated with 0.9% sodium chloride injection. These findings indicate disproportionate reporting within FAERS and should be considered candidate safety signals rather than evidence of causality. The high disproportionality values for reactions with relatively small numbers of reports, warrant further clinical case review and assessment of concomitant medications, underlying conditions, and potential reporting biases. The heatmap summarizes PRR and Chi-square values for detected signals, while the MA plot illustrates the relationship between drug-ADR reporting abundance and log-transformed PRR.

3. Time-Trend Analysis

FAERS report dates were converted to datetime format and aggregated by month for Q3 2020 (July–September 2020). Unique PRIMARYID values were used to ensure that multiple drug or adverse-reaction records belonging to the same report were not counted as separate reports.

The analysis showed that total reporting volume decreased from July to August and increased again in September. Reports involving the most frequently reported drug, ZANTAC, decreased across the three-month period, while reports associated with the top adverse reaction, Drug ineffective, decreased in August and increased again in September.

Because FAERS is a spontaneous reporting system, these changes represent patterns in reporting frequency rather than changes in the true incidence or risk of an adverse event. Differences in

reporting may also be influenced by factors such as exposure, reporting practices, and other characteristics of spontaneous reporting.

4. Limitations

This analysis has several important limitations. One is FAERS is a spontaneous reporting database and therefore does not provide a complete population denominator. Consequently, disproportionality measures describe relative reporting patterns within FAERS, rather than absolute incidence or individual patient risk. FAERS reports may also contain duplicate information, incomplete demographic data, multiple suspected or concomitant drugs, and reporting biases. Factors such as media attention, regulatory warnings, stimulated reporting, and differences in reporting practices can influence observed reporting frequencies. Most importantly, a statistical signal does not establish causality. A detected drug-ADR association requires further clinical and epidemiological evaluation. Rare combinations with small cell counts can also produce unstable or extreme estimates. In particular, zero cells can result in infinite PRR or undefined ROR estimates.

5. Next Step:

- Validate candidate signals using real-world observational datasets (e.g., Sentinel, OMOP Common Data Model) with self-controlled case series (SCCS)
- Implement the Multi-item Gamma Poisson Shrinker (EBGM) to stabilize variance in ultra-rare events and protect against small-count false alarms.
- Deploy monthly time-series monitoring using changepoint detection algorithms to flag accelerated reporting velocity across newly approved biological agents.
- Rerun disproportionality queries stratified by patient sex, age bracket, and reporting country to eliminate masking effects and regional reporting bias.